

ID	Requirement	Final description / acceptance criterion	Priority
FR1	Dashboard Homepage	Show class overview, High/Medium/Low risk totals, key alerts and summary insights.	Must
FR2	Student List	Display students and their current risk, key metrics and status.	Must
FR3	Student Profile	Dedicated profile containing attendance, assessment, behaviour, homework, trends, risk and	Must
FR4	Attendance Analysis	Analyse current attendance and change over time , identifying deteriorating patterns.	Must
FR5	Assessment Analysis	Analyse achievement across assessment periods and identify decline/improvement.	Must
FR6	Behaviour Analysis	Analyse incidents/points and repeated or changing behaviour patterns.	Should
FR7	Homework Analysis	Analyse completion and identify repeated missing-work or declining-completion patterns.	Should
FR8	At-Risk Identification	Automatically flag students likely to need additional support.	Must
FR9	Risk Classification	Produce Low/Medium/High risk categories and, where appropriate, predicted probabilities.	Must
FR10	Explainability Engine	Explain in plain English why the student was classified at a particular risk level.	Must
FR11	Contributing-Factor Analysis	Rank or identify the most important factors contributing to each prediction.	Must
FR12	Student Summary	Generate concise evidence-based summaries from analysed data.	Must
FR13	Personalised Recommendation Engine	Recommend interventions/support actions based on the student's identified risk factors rather than	Must
FR14	Recommendation Explanation	Explain why each recommended action has been suggested.	Must
FR15	Parent Email Draft	Generate an editable draft based on the student's actual concerns and proposed support.	Should
FR16	Dashboard Visualisations	Charts for trends, class distribution and relevant student metrics.	Must
FR17	Search and Filter	Search student names and filter by class/risk category.	Should
FR18	Intervention Tracking	Record interventions and their status/outcomes.	Could
FR19	Export Student Summary	Export a student summary for meetings/records.	Could
FR20	Class-Based Data Upload	Upload data for a selected class through the application.	Must
FR21	Attendance Upload	Upload attendance CSV using documented structure.	Must
FR22	Assessment Upload	Upload assessment records, including multiple assessment periods.	Must
FR23	Behaviour Upload	Upload behaviour records.	Should
FR24	Homework Upload	Upload homework records.	Should

FR25	Download Data Templates	Provide valid CSV templates for each data source.	Must
FR26	Data Validation	Detect missing columns, invalid types, impossible values, duplicates and missing data.	Must
FR27	Data Cleaning	Clean/transform uploaded data before analysis and document transformations.	Must
FR28	Missing-Data Handling	Use an explicit, documented strategy rather than silently ignoring missing records.	Must
FR29	Longitudinal Data	Support multiple periods so trends can be calculated rather than relying solely on snapshots.	Must
FR30	Dataset Documentation	Document variables, units, target, provenance, limitations and ethical handling of the dataset.	Must
DS1	Feature Engineering	Derive useful features such as assessment change, attendance change, homework trend and behaviour	Must
DS2	Prediction Target	Define a clear and defensible prediction target, e.g. At Risk / Not At Risk, before model training.	Must
DS3	Baseline Model	Retain the original transparent rule-based risk engine as a baseline comparator.	Must
DS4	Logistic Regression	Train and evaluate Logistic Regression as an interpretable ML baseline.	Must
DS5	Decision Tree	Train and evaluate a Decision Tree classifier.	Must
DS6	Random Forest	Train and evaluate a Random Forest classifier.	Must
DS7	Train/Test Separation	Keep testing data separate from model training.	Must
DS8	Cross-Validation	Use k-fold/stratified cross-validation when appropriate and report the results.	Must
DS9	Class-Imbalance Analysis	Check whether risk classes are imbalanced and justify any treatment used.	Must
DS10	Multiple Evaluation Metrics	Report accuracy, precision, recall and F1; include ROC-AUC for appropriate binary classification.	Must
DS11	Confusion Matrix	Show where the chosen model makes false-positive and false-negative predictions.	Must
DS12	ROC / Precision-Recall Visualisation	Generate relevant performance plots for the final model.	Must
DS13	Model Comparison	Compare the rule baseline, Logistic Regression, Decision Tree and Random Forest in one results	Must
DS14	Model Selection	Select the production model based on evidence, not merely whichever has the highest accuracy.	Must
DS15	Statistical Comparison	Include an appropriate statistical or repeated-CV comparison where methodologically defensible.	Must for 70%+
DS16	Model Explainability	Analyse model coefficients, feature importance or another appropriate interpretable method.	Must
DS17	Recommendation Evaluation	Evaluate whether recommendations correspond logically to detected risk factors and teacher needs.	Must
DS18	Error Analysis	Investigate examples the model classifies incorrectly and explain probable reasons.	Must
DS19	Robustness Testing	Test behaviour with missing, unusual, boundary and altered input data.	Must

DS20	Reproducible Pipeline	Separate preprocessing/training/evaluation sufficiently that experiments can be rerun	Must
UI1	Basic Teacher Login	Simple access/login screen before entering Phronesis. No enterprise authentication required.	Must, low effort
UI2	Navigation	Clear navigation between Dashboard, Students, Upload, Analysis and Settings.	Must
UI3	Settings	Small settings screen for relevant configurable options only.	Should
NFR1	Usability	Teacher should be able to understand the main workflow without instructions.	Must
NFR2	Low Cognitive Load	Prioritise alerts, summaries and explanations rather than excessive charts/data.	Must
NFR3	Explainability	Every risk prediction should be accompanied by supporting evidence.	Must
NFR4	Privacy	Use anonymised/synthetic/public research data and do not expose identifiable real pupils.	Must
NFR5	Accessibility	Do not rely only on red/amber/green; use text labels, clear typography and readable layouts.	Must
NFR6	Reliability	Identical input/model state should generate consistent outputs.	Must
NFR7	Maintainability	Separate data processing, ML, recommendations and UI logic into sensible modules.	Should
NFR8	Performance	Process normal class-sized uploaded datasets without unreasonable delay.	Should
NFR9	Reproducibility	Record model/configuration/random seeds/version dependencies where relevant.	Must
NFR10	Error Handling	Fail gracefully when the wrong data/file structure is supplied.	Must

Why it matters
Core usable product
Core teacher workflow
Complete teacher decision-support flow
Genuine data analysis
Genuine data analysis
Adds richer multidimensional
Additional predictive indicator
Central Phronesis output
Core analytical output
XAI / teacher trust
Makes ML actionable
Supports rapid interpretation
Aligns Phronesis with recommendation-
Strengthens explainability and
Product completeness
Template explicitly expects visualisation
Usability
Good extension but not core
Useful extension
Data ingestion
Data ingestion
Required for longitudinal analysis
Richer feature set
Richer feature set

Reliability/usability
Data-quality requirement
Explicit Data Science pipeline requirement
Methodological quality
Stronger analytics
High-quality dataset evidence
First/2:1 Data Science criterion
Prevents vague ML
First-class baseline comparison
Model comparison
Model comparison
More advanced ML approach
Valid evaluation
Explicit template criterion
Stronger methodology
Template expects multi-metric
Critical model analysis
Directly aligns with template outputs
First-class evidence
Critical evaluation
Outstanding-template criterion
Connects ML to Phronesis XAI
Template expects meaningful
Demonstrates critical evaluation
Reliability/evaluation

Professional Data Science practice
Makes application coherent
Finished-product quality
Matches design without feature creep
User-centred design
Existing research/design
Human-centred AI
Ethics
Inclusive design
Trust
Code quality
Product usability
Strong experimental practice
Professional-quality application